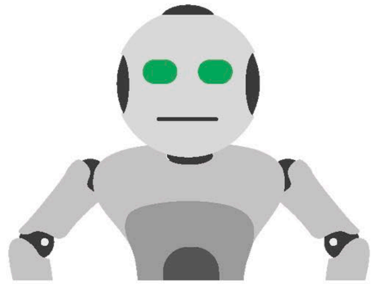
Challenges

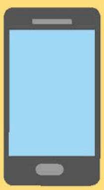
Before making predictions about the future, it's important to have a detailed understanding of current science. This limits the number of people who can make educated guesses. On top of that, the advancement of technology is a collaborative effort, where researchers, politicians, users, and businesses all have a part to play. It's almost impossible to see all the pieces at once.

△ **Timing**
Research often takes decades to generate results. The breakthrough could occur in one year, a few years, or not at all. No one knows for sure.

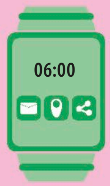
△ **Social behaviours**
Companies didn't expect smartphones to be used as flashlights. Even the most far-sighted developers can't predict how people will respond to a new technology.



△ **Rise of interdisciplinary fields**
Fields like robotics or neuroscience, which draw from multiple fields, are extra tricky to predict because few people fully understand all their complexities in depth.



△ **Mobile phones**
Phones can be used for everything from taking pictures to streaming music. This has created a feedback loop where industries create even more mobile content, which leads to people using their phones even more.



△ **Wearable tech**
Golf shoes that analyse your game, rings that can act as credit cards, fitness bracelets, and glasses with AR displays are already available to buy. These devices can link together and share information.

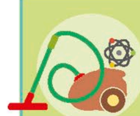
△ **Cryptocurrencies**
Cryptocurrencies are digital currencies that use cryptography to make financial transactions easier and more secure. The first, Bitcoin, was released in 2009. The value of a cryptocurrency can go up or down unpredictably.

△ **Digital assistants**
Advances in speech recognition and natural language processing (NLP) have allowed AI to be placed on devices as small as phones and watches. Navigating the digital world has now become easier than ever.

REAL WORLD

Failed predictions

New discoveries are exciting, but often poorly understood. Whether it's electricity, artificial intelligence, or quantum computing, every big leap in science is accompanied by a flood of bad predictions. Other discoveries are way ahead of their time, and require decades of research to develop the necessary supporting technology to get the product on the shelves. Below are some spectacular failed predictions.



Nuclear-powered vacuum cleaners (1955)

Missile-guided mail delivery (1959)

Vacations to the Moon (1969)